Calculation of Tractive force in 2nd Gear

$Loaded\ Radius\ of\ Wheel\ = 450\ mm$

$Torque\ from\ engine\ in\ Nm\ at\ 1500\ rpm\ = 160\ Nm$

$Engine\ inertia\ = 0.0904\ \text{kgm}^2$

$Transmission\ inertia\ in\ 2nd\ gear\ = 0.102\ \text{kgm}^2$

$Transmission\ in\ 2nd\ gear\ ratio\ = 2.79$

$Overall\ drive\ system\ efficiency\ = 0.960$

$Driveline\ \ inertia\ = 0.136\ \text{kgm}^2$

$Wheel\ \ inertia\ = 1.24\ \text{kgm}^2$

$Driveline\ \ Ratio\ = 2.92$

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$Acceleration\ of\ car\ \ when\ switching\ from\ 1st\ to\ 2nd\ gear = 1.3\ m/s^2$

$$\boldsymbol{N_{tf} = N_t N_f}$$

$$\boldsymbol{N_{tf} = 2.79 * 2.92}$$

$$\boldsymbol{N_{tf} = 8.15}$$

$$\boxed{(\boldsymbol{F_x}) = \frac{\boldsymbol{T_e N_{tf} \eta_{tf}}}{\boldsymbol{r}} - \{(\boldsymbol{I_e + I_t})\boldsymbol{N_{tf}^2} + \boldsymbol{I_d N_f^2} + \boldsymbol{I_w}\}\frac{\boldsymbol{a_x}}{\boldsymbol{r^2}}}$$

$N_{tf} = Transmission\ gear\ ratio\ * Differential\ gear\ ratio$

$T_e = Torque\ from\ engine\ in\ Nm$

$I_e = rotational\ inertia\ of\ engine\ in\ kgm^2$

$I_t = rotational\ inertia\ of\ transmission\ in\ kgm^2$

$I_d = rotational\ inertia\ of\ differential\ in\ kgm^2$

$I_w = rotational\ inertia\ of\ wheel\ in\ kgm^2$

$F_x = traction\ force$

$r = radius\ of\ wheel$

$a_x = linear\ acceleration\ of\ vehicle\ or\ wheel$

$$\boxed{(\boldsymbol{F_x}) = \frac{\boldsymbol{160 * 8.15}}{\boldsymbol{0.45}} - \{(\underbrace{\boldsymbol{0.0904 + 0.145)66.42\ + 0.136\ * 8.53 + 1.24}}\}\frac{\boldsymbol{1.3}}{\boldsymbol{0.45^2}}}$$

Effective Inertia = 18.03 kgm^2

$$\boxed{(\boldsymbol{F_x}) = \boldsymbol{2898 - 115.7}}$$

$$\boxed{(\boldsymbol{F_x}) = \boldsymbol{2782.3\ N}}$$

$\eta_{tf} = Overall\ drive\ system\ efficiency$

Calculation of Traction force in 2nd Gear

A car is accelerating, and the driver has shifted from 1st gear to 2nd gear .
The max rpm at 1st gear is 1500 rpm and Torque is 120 Nm
The vehicle is at 20 kmph having covered reached from 0 in 3 seconds .
What is the total traction force available at the Wheel ends ?

$$(F_x) = \frac{T_e N_{tf}}{r} - \{I_{effective}\}\frac{a_x}{r^2}$$

$$I_{effective} = 37\ kgm^2$$

$$T_e = 120\ Nm$$

$$Transmission\ in\ 2nd\ gear\ ratio, N_t = 2.5$$

$$Driveline\ \ Ratio, N_f = 3$$

$$Loaded\ Radius\ of\ Wheel. r\ \ = 400\ mm$$

$$Instantaneous\ acceleration\ of\ the\ vehicle\ when\ shifting, a_x = \frac{v-u}{t} = \frac{5.556-0}{3} = 1.852\ ms^{-2}$$

Find the Tractive Force generated at the Wheel to maintain the acceleration overcoming rotational inertia

$$\boxed{(F_x) = \frac{120 * 7.5}{0.4} - \{37\}\frac{1.852}{0.4^2}}$$

$$\boxed{(F_x) = 1821.725}$$